

K- Mean Example

Given { 2,4,10,12,3,20,30,11,25}
Form 2 clusters

- Randomly assign means $m_1 = 3$, $m_2 = 4$
- The numbers which are close to m_1 are grouped into cluster K_1 and numbers which are close to m_2 are grouped into cluster K_2 .
- Again calculate mean for new cluster groups
- Iteration 1
- $K_1 = \{ 2, 3\}$ $K_2 = \{ 4,10,12,20,30,11,25\}$
- $M_1 = 2.5$ $M_2 = 16$

- Iteration 2

$K1 = \{2, 3, 4\}$ $K2 = \{10, 12, 20, 30, 11, 25\}$

$M1 = 3$ $M2 = 18$

- Iteration 3

$K1 = \{2, 3, 4, 10\}$ $K2 = \{12, 20, 30, 11, 25\}$

$M1 = 4.75$ $M2 = 19.6$

- Iteration 4

$K1 = \{2, 3, 4, 10, 11, 12\}$ $K2 = \{20, 30, 25\}$

$M1 = 7$ $M2 = 25$

- Iteration 5

$K1 = \{2, 3, 4, 10, 11, 12\}$ $K2 = \{20, 30, 25\}$

$M1 = 7$ $M2 = 25$

Stop as the clusters with these means are the same in iteration 4 and iteration 5.

- So the final cluster is
- $K1 = \{ 2,3,4,10,11,12\}$
- $K2 = \{20,30,25\}$